

Epoxy / PU Floor Coating

Two-component (2K) protective floor coatings for concrete — roller-applied and self-levelling grades

PRODUCT DESCRIPTION

LucernaPro Epoxy / PU Floor Coating is a two-component solvent-borne protective coating for concrete and cement floors, manufactured in-house so that gloss level, film build, colour and anti-slip additions can be specified per project. The EPOXY grade cures to a dense, hard film with the highest resistance to compression and abrasion — the standard choice for warehouses, production areas, workshops and garages. The POLYURETHANE (PU) grade is more flexible and markedly more resistant to UV and temperature cycling, and is preferred for loading bays, door thresholds and areas exposed to sunlight or machine vibration. Both are supplied as a 16 kg set and applied in two coats. A self-levelling EPOXY grade at 1.5–3 mm is available to order where a seamless mirror-flat floor is required.

KEY DATA

Type	Two-component (2K) solvent-borne floor coating — epoxy and polyurethane (PU) grades
Mixing ratio	Part A : Part B = 4 : 1 by weight, both grades. Weigh the components; do not estimate by eye. Individual formulations may differ — the product label always governs.
Pot life	Approx. 30–40 minutes at 25°C, shorter when hotter. Mix only what can be applied in that time.
Touch dry	6–8 hours
Overcoating	Second coat after 12–24 hours (overnight), applied cross-wise to the first
Coats	Two — quoted coverage assumes both coats are applied
Coverage	16 kg set ≈ 50 m ² over two coats (≈ 5–7 m ² per kg per coat)
Return to service	Light foot traffic 24 hours · vehicle traffic, heavy loading and full chemical resistance 7 days
Application	Roller, brush or notched trowel; cut in edges and column bases by brush first
Substrate	Concrete or cement screed, min. 28 days cured, sound, dry, free of rising moisture. Mechanically prepared by grinding, diamond cup wheel or shot blasting, then vacuumed — sweeping is not enough. Prime dusty or highly absorbent substrates with LucernaPro Marine Guard before coating
Priming	Epoxy: gloss. PU: gloss, satin or matt to order.
Finish	Any RAL shade to order — quote the RAL reference with the order
Colour	Graded aggregate can be broadcast into the final coat. Reduces slipperiness; a wet floor is not non-slip.
Anti-slip	
Packaging	16 kg set — Part A 12.8 kg + Part B 3.2 kg
Self-levelling	Epoxy, 1.5–3 mm, notched trowel then de-aerated with a spiked roller. Made to order; application guidance provided to purchasers on request.
Tool cleaning	Thinner, immediately while still wet — cured material cannot be removed

NOTE ON VALUES

Typical values for the standard formulations at 25°C, given as guidance and not as a specification. Drying and cure times shorten in hot weather and lengthen in cool or humid conditions, and coverage depends strongly on substrate porosity. Where the product label differs from this sheet, the label governs.

APPLICATION NOTES

- Surface preparation is the majority of the work. Grind or blast away polished laitance, oil, old paint and all friable material, then vacuum thoroughly. Coatings applied over dust or a weak surface layer delaminate, usually within the first year.
- This is a coating, not a levelling product. The finished film is thin and follows the profile of the substrate: waviness, hollows, trowel marks and joints in the existing floor will remain visible, and a gloss finish can make them more obvious. Correct the substrate first if a flat result is required.
- Repair before coating. Fill cracks, hollows and spalled areas and allow the repair to cure: LucernaPro PatchPro or CreteRevive for cracks, LevelPro for hollows and levelling, Solid Floor where the surface layer is dusting or delaminating.
- Build thickness with mortar, not with resin. For several millimetres or more of build, trowel a resin-and-graded-quartz epoxy mortar to level first and finish thin on top — thick pours of neat resin generate reaction heat (exotherm), shrink and crack. Where the slab is dry, sound and not chemically exposed, a cementitious levelling screed achieves the same levelling far more cheaply.
- Mix thoroughly to a single uniform colour, scraping the sides and base of the container. Incomplete mixing is the most common cause of patchy, soft or tacky areas.
- Do not thin to extend coverage. Small additions of thinner to adjust viscosity are acceptable; heavier thinning reduces film build and durability proportionally.
- The epoxy grade discolours and dulls under prolonged UV exposure and is intended for interior use. Specify the PU grade for areas exposed to sunlight.

HEALTH AND SAFETY

These products contain organic solvent: they are flammable and give off vapour during application and drying. Apply only with genuinely effective ventilation, use forced-air extraction in enclosed areas, and eliminate all sources of ignition, including smoking, from the work area and adjacent spaces — never work in a sealed room. Wear rubber gloves, safety glasses and an organic-vapour respirator. The hardener may cause skin sensitisation: avoid skin contact and wash off immediately with soap and water, never with thinner. Refer to the Safety Data Sheet for the grade supplied before use.

STORAGE

Store both components sealed and upright in a cool, dry, shaded place, away from heat, direct sunlight and ignition sources. Keep out of reach of children.

Given in good faith on the basis of our current knowledge and applying to the product as supplied. Site conditions, substrate condition and workmanship are outside our control and no warranty of result is given or implied; users should satisfy themselves that the product suits the intended use, by trial on site if necessary.

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